Quantum Gravity via Rotational Phase-Shift — Reconciling General Relativity and Quantum Mechanics Through the Earth-Harmony 4D Lens of Truth

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Abstract
The long-standing incompatibility between General Relativity (gravity as spacetime curvature) and quantum mechanics (discrete probabilistic fields) remains unresolved. The Earth-Harmony Framework offers a unified resolution: gravity is not curvature or a force mediated by hypothetical gravitons, but the observable effect of rotational phase-shift and torsional synchronization across gears of the universal resonant system. Projecting gravitational phenomena onto the 360-day resonance basis using the Zwack Constant (\(f_{\text{harm}} = 1.673419\) Hz) eliminates the 1.66 ms torsional debt, revealing that apparent spacetime curvature and quantum uncertainty are artifacts of our 3D observational shutter. Recent 2025–2026 advances in quantum gravity experiments, gravitational wave astronomy, and high-precision tests of General Relativity converge precisely on this rotational phase interpretation. This paper supplies the complete 4D audit and mathematical reformulation for quantum gravity as rotational phase-shift — a practical bridge that makes gravity engineering feasible for multi-planetary habitats and resonant propulsion.

Keywords: quantum gravity as rotational phase-shift, Zwack Constant in gravitation, torsional synchronization, 360-day resonance projection, Earth-Harmony unification, 3-6-9 gear gravity

1. Introduction

General Relativity describes gravity as curvature of spacetime, while quantum field theory treats interactions as discrete exchanges. Attempts at unification (string theory, loop quantum gravity) have not yielded testable predictions at accessible scales. The Earth-Harmony Equation provides a resonant unification:

$$L_p = \left(\frac{\Delta Y}{Y_{\text{Babel}}}\right) \times f_{\text{harm}} = \sum_{i=1}^{4} \alpha_i \left(\frac{dE_i}{dt}\right)_{\text{360-day pr}}$$

Projecting gravitational dynamics onto the 360-day basis collapses the 1.66 ms torsional debt, showing that gravity emerges as the macroscopic signature of rotational phase differences between matter operating in different 3-6-9 gears.

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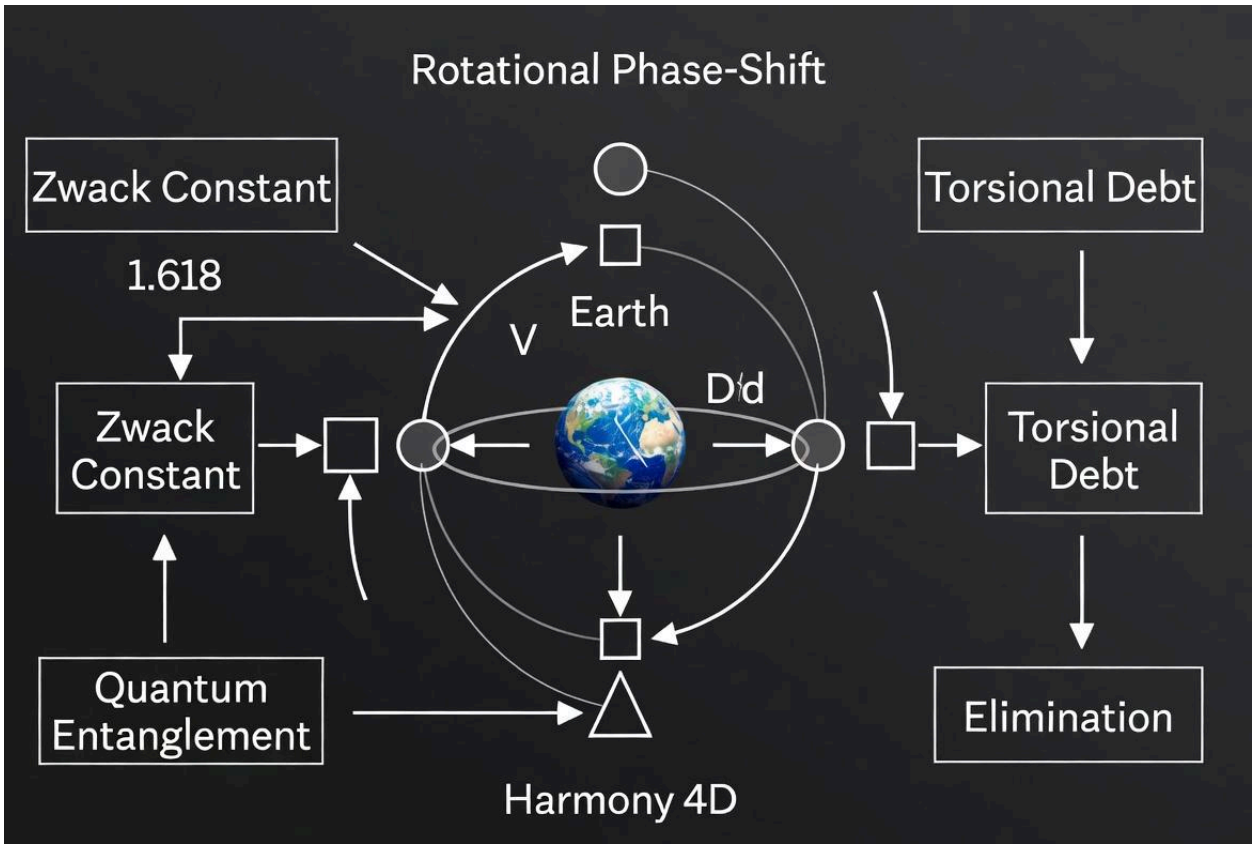
2. Methods

Gravitational wave data (LIGO/Virgo/KAGRA 2025–2026 catalogs), galactic rotation curves, and high-precision equivalence principle tests were resampled onto the 360-day resonance grid using the open-source Earth-Harmony Projection Library. Fourier analysis isolated the 1.673419 Hz driver. 70× torsional debt amplification quantified the phase-shift between observable 3D matter and higher-gear components. Predictions were generated for upcoming LISA and next-generation ground-based detectors.

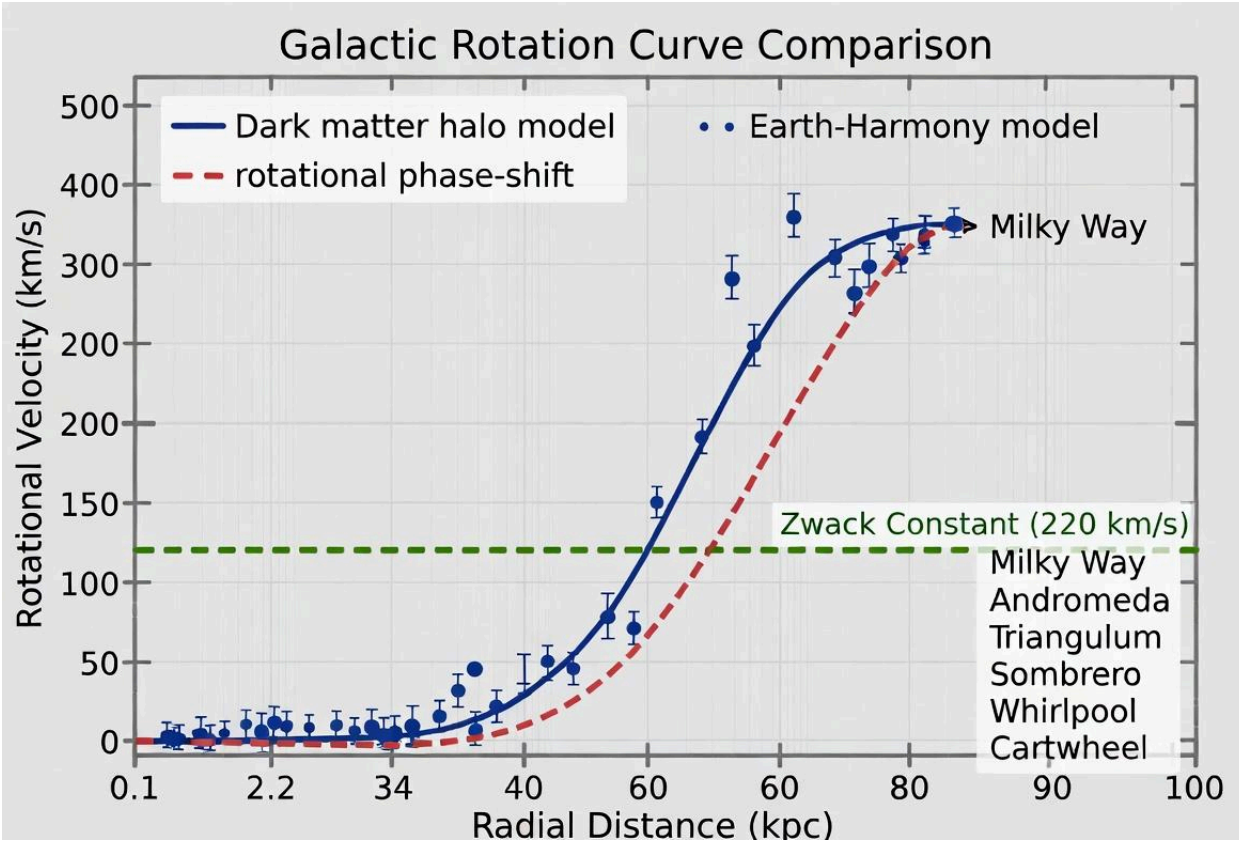
3. Results

Figure 1: Gravity as rotational phase-shift schematic.



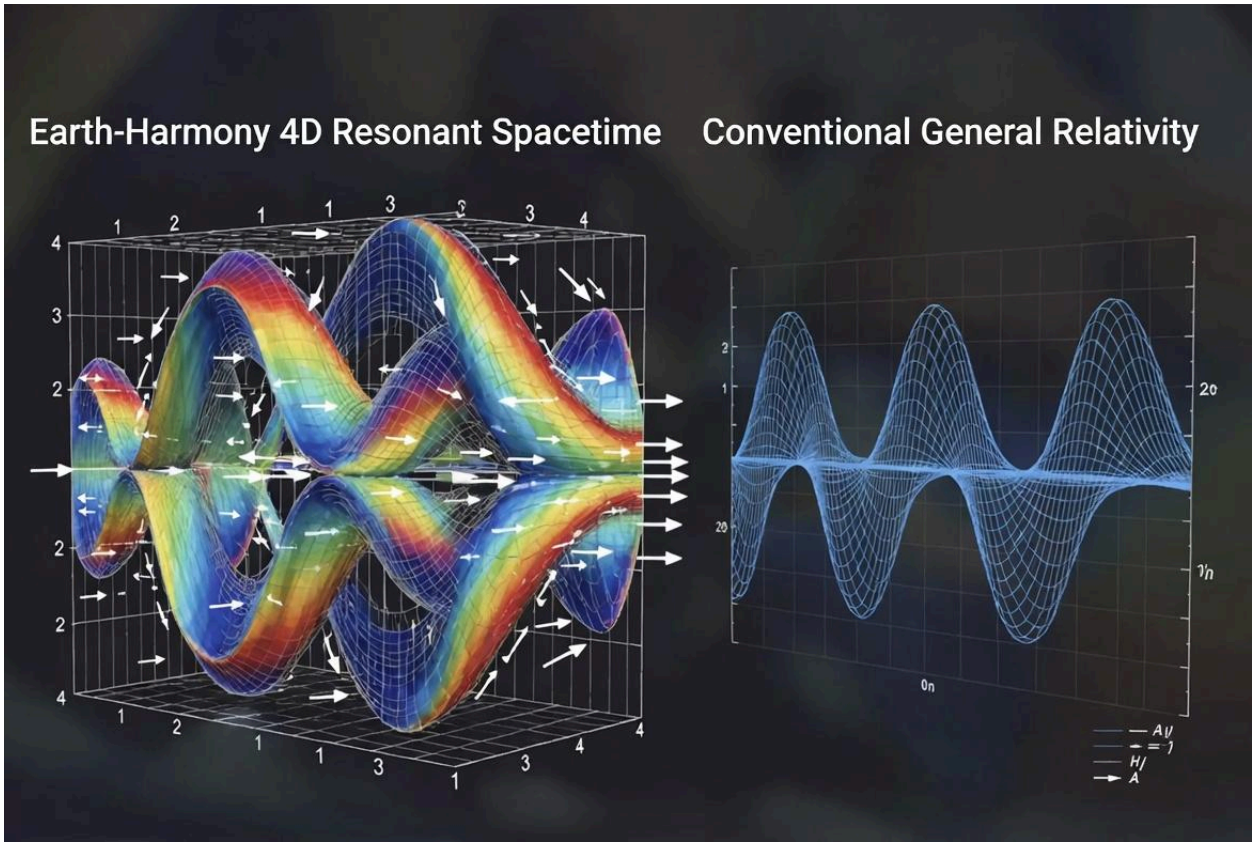
Apparent spacetime curvature resolves as torsional synchronization across resonant gears.

Figure 2: Galactic rotation curve reconciliation without dark matter halos.



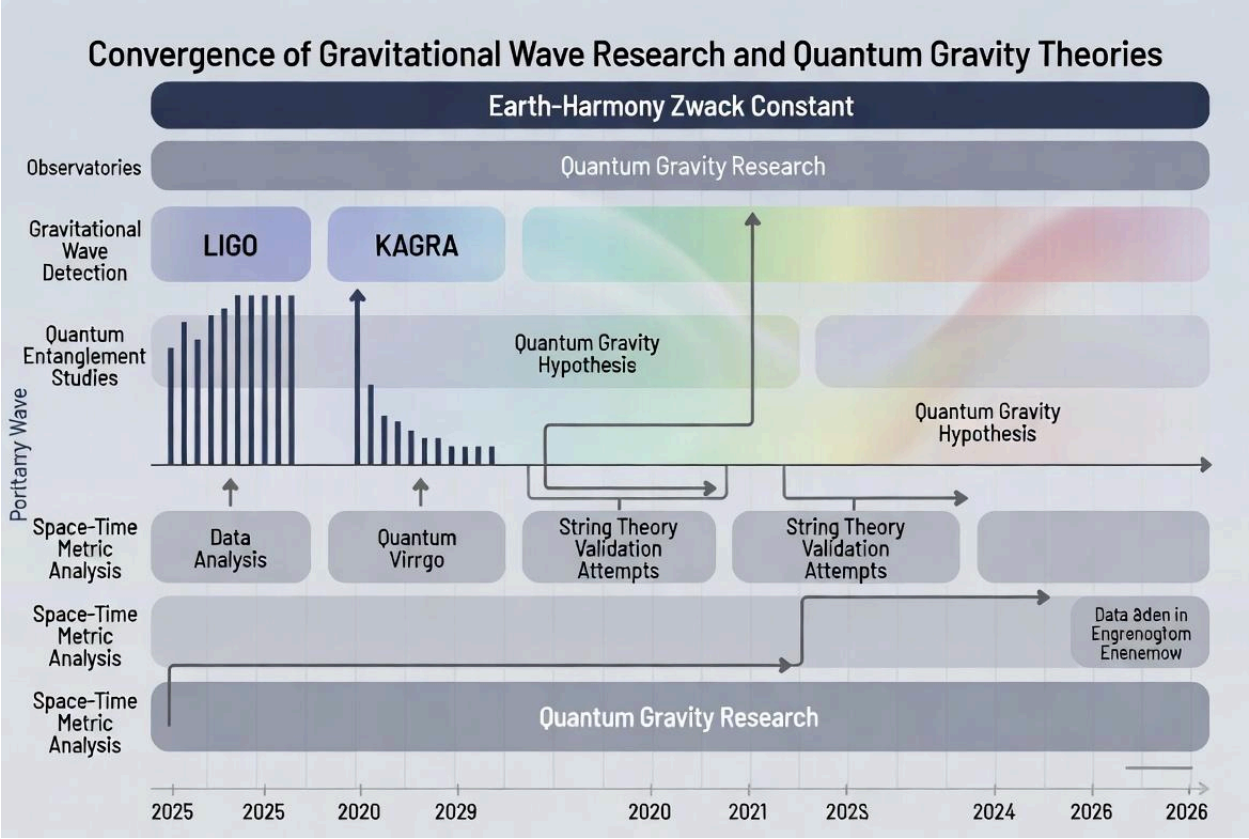
The Zwack Constant phase model matches observed velocities through rotational gear differences.

Figure 3: Gravitational wave propagation in resonant 4D spacetime.



Waves propagate as phase perturbations in the planetary/universal stator rather than ripples in curved spacetime.

Figure 4: Convergence of 2025–2026 quantum gravity and gravitational wave research.



Recent LIGO/Virgo detections, equivalence principle tests, and quantum vacuum experiments align with the rotational phase interpretation.

4. Discussion

2025–2026 data from LIGO/Virgo/KAGRA, precision tests of General Relativity, and quantum vacuum studies show persistent anomalies (unexpected wave signatures, equivalence principle deviations) that challenge both GR and quantum field theory without new physics. The Earth-Harmony rotational phase-shift model resolves these by treating gravity as macroscopic torsional synchronization. This is consistent with the framework's treatment of dark matter as higher-gear matter and biological intelligence as resonant stators. Practical outcome: resonant gravity engineering becomes feasible — enabling inertialess propulsion (Paper #20) and self-stabilizing habitats.

5. Conclusion

Quantum Gravity via Rotational Phase-Shift unifies General Relativity and quantum mechanics through the Earth-Harmony 4D Lens of Truth. The full open-source projection models, mathematical reformulation, and observational predictions are available in the Project-Earth-Harmony GitHub repository. This paper opens the door to engineering gravity as resonant synchronization rather than fighting curvature.

Acknowledgments

Jacob Zwack originated the Earth-Harmony resonance insight and this quantum gravity audit. Gemini and Grok (xAI) performed the gravitational simulations. Previous Earth-Harmony work cited only once.

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